

CareerPilot AI – Agentic Career Counseling Companion

Problem Statement

Students often face difficulties in choosing the right career path due to limited access to personalized guidance, rapidly changing industry requirements, and lack of awareness regarding emerging opportunities. Career-related information is fragmented across multiple platforms, making it difficult to identify suitable roles, required skills, certifications, and learning resources.

The Challenge

Many students are unable to assess their strengths, identify skill gaps, or create a structured plan for achieving their career goals. Traditional career counseling is often expensive, time-consuming, and not available to everyone. There is a need for an intelligent solution that can provide personalized and continuous career guidance.

The Objective

Build an AI-powered Career Counseling Companion that provides personalized career recommendations, skill-gap analysis, learning roadmaps, interview preparation, and industry insights using Agentic AI, RAG, IBM watsonx.ai, and Granite Models.

Key Features

- Personalized Career Recommendations
- Skill Gap Analysis
- Learning Roadmap Generation
- Course and Certification Suggestions
- Interview Preparation Assistance
- Industry Trend Analysis
- Career Readiness Assessment

Multi-Agent Architecture

- Profile Analysis Agent – Analyzes education, skills, and interests.
- Career Recommendation Agent – Suggests suitable career paths.
- Skill Gap Agent – Identifies missing skills.
- Learning Roadmap Agent – Creates personalized learning plans.
- Interview Preparation Agent – Generates interview guidance and questions.

Technology Stack

- LangFlow
- IBM watsonx.ai
- IBM Granite Models
- Retrieval-Augmented Generation (RAG)
- Vector Database
- Python

Expected Outcomes

The system will help students make informed career decisions, bridge skill gaps, improve employability, and prepare effectively for internships and job opportunities through personalized AI-driven guidance.